6 Education and training

About WHMIS education and training for workers

In brief

- Employers must educate and train workers on the hazards and safe use, handling, and storage of hazardous products.
- Employers are required to educate and train workers about WHMIS 2015 (new pictograms, signal words, hazard classes, labels, SDSs, etc.).
- Workers must participate in WHMIS training programs, take necessary steps to protect themselves and their co-workers, and participate in identifying and controlling hazards.

Under WHMIS 2015, education and training can be thought of as two separate parts:

- **Education** refers to general information such as:
  - How WHMIS works
  - The major hazards of the hazardous products in use in the workplace
  - The rights and responsibilities of employers and workers
  - The content required on labels and SDSs, and the significance of this information
- **Training** refers to the site- and job-specific information to employees that will cover your workplace's procedures for storage, handling, use, disposal, emergencies, spills, and what to do in unusual situations.

Who should receive education and training?

If a workplace uses hazardous products, a WHMIS program must be in place. Workers must be educated and trained so they understand the hazards and know how to work safely with hazardous products.

All workers who work with a hazardous product, or who may be exposed to a hazardous product as part of their work activities, must learn about the hazard information for these products. (For example, a receptionist at a dental office may be exposed to mercury if it is spilled.) The hazard information should include the information received from the supplier, as well as any other information that the employer is aware of about the use, storage, and handling of each product.

As an example, this education and training will include all workers who:

- May be exposed to a hazardous product due to their work activities (including normal use, maintenance activities, or emergencies)
- Use, store, handle, or dispose of a hazardous product
- Supervise or manage workers who may be exposed, or use, store, handle, or dispose of a hazardous product
- Are involved in emergency response

Who should provide the education and training?

WHMIS legislation places the obligation for education and training with the employer, and it outlines the minimum requirements for education and training. This education and training may be provided by the employer, or by a qualified person or agency chosen by the employer. Regardless of who delivers the education and training, employers remain legally responsible to ensure the protection of workers.

Topics to be covered

Examples of topics that should be covered during education and training include:

- The information on both supplier labels and workplace labels, and what that information means
- The information on the safety data sheet (SDS) and what that information means
- The procedures required for safe use, handling, and disposal of a hazardous product
- Any other information required when the product is in a pipe, piping system, vessel, tank car, etc.
- Procedures to follow if the hazardous product may be present in the air and a worker may be exposed
- All procedures that must be followed in an emergency that involves the hazardous product

Duties for education and training

Employers

Employers must develop, implement, and maintain a worker WHMIS education and training program. Education and training is required for workers who may be exposed to hazardous products that are produced or used at the workplace.

The employer has the general responsibility to provide workers with all of the hazard information possible, either from the supplier or based on information the employer is, or ought to be, aware of.

Employers are expected to consult with the health and safety committee (or representative) when developing, implementing, or reviewing the education and training programs.

In addition, employers must review their overall WHMIS education and training programs at least annually, or more often if there is a change in work conditions, hazard information, or similar factors. This review should be done in consultation with the health and safety committee or representative.

Refresher education and training is generally required:

(1) As needed to protect workers' health and safety
(2) If conditions of the workplace have changed
(3) If new products are introduced
(4) If the products have changed and now have different hazards
(5) When new hazard information becomes available
(6) If there is new information about safe use, handling, storage, or disposal

Employers must periodically evaluate workers' knowledge using written tests, practical demonstrations, and other suitable means.

Workers

Workers must participate in the education and training sessions. They must also follow the safe work procedures established by their employer.

Workers who are successfully educated and trained in WHMIS must be able to answer these four key questions for every hazardous product they work with:

- What are the hazards of the product?
- How do I protect myself from those hazards?
- What do I do in case of an emergency?
- Where can I get further information?

When the education and training program is reviewed, it is important for employers to find out if workers still understand the hazards of hazardous products and follow safe work procedures. These four key questions can also be used to evaluate whether workers need to be retrained because they have forgotten some information.

Does WorkSafeBC validate education or training programs?

WorkSafeBC does not validate education or training programs. Private WHMIS consultants can help you with education or training. WorkSafeBC prevention officers will conduct performance-based audits as part of their workplace inspections. For example, they may evaluate workers' knowledge of health and safety information specific to the products they work with or near.

External companies that use high-pressure sales tactics have contacted some employers. Keep in mind that employers have a choice when deciding on an external education or training provider.

Some industries, such as construction, offer WHMIS "cards" or "certificates" to participants who complete their WHMIS education program. Such cards and certificates can be useful for workers who move regularly from site to site, enabling them to prove to new employers that they have attended WHMIS sessions. However, presentation of a card, for example, does not necessarily guarantee that a worker understands and can apply the information. Unless written evaluation has taken place, cards and certificates merely document attendance. **Job-specific training at each worksite is still required for all workers who work with or near hazardous products.**

Implementing WHMIS 2015 in your workplace

What is an exposure control plan?

An exposure control plan (ECP) is a requirement of WorkSafeBC's Occupational Health and Safety Regulation. An ECP sets out a detailed approach to protecting workers from harmful exposure to designated substances (for example, carcinogens, sensitizers, and reproductive toxins) and, under certain conditions, for all other hazardous products.

To implement the WHMIS program, employers should make use of supplier labels and SDSs, as well as their own knowledge of the hazards of products and their use in the workplace. Their workplace knowledge should take into account factors such as quantity, work processes, control measures, and work location. For example, the hazards of spray painting with a hazardous product inside a confined space are far different from the hazards of hand brushing the same product outdoors.

Based on all this information, employers must develop exposure control plans (when required) and written safe work procedures that ensure the health and safety of workers. They must also educate their workers about the hazards and train them in safe work procedures.

How employers implement WHMIS in their own workplaces will vary, but the major elements of their programs will be similar. The "WHMIS 2015 implementation plan checklist" on page 83 can be used to verify that a workplace WHMIS implementation plan meets WorkSafeBC guidelines. In developing a program to instruct workers, an employer might use the "WHMIS 2015 education and training checklist" on page 86.

What are safe work procedures?

Employers must develop effective procedures to prevent exposing their workers to excessive levels of hazardous products. These include safe work procedures for handling, using, storing, and disposing of hazardous products. As well, safe work procedures must be established for emergency situations and spill cleanups.

Safe work procedures should be written, and they should address the specific hazards of the hazardous product and how it is used in the workplace. They must contain enough detail to provide direction to workers. Workers must be trained by the employer and must follow these procedures at the worksite.

Sample safe work procedure

The following is an example of a written safe work procedure to be used by authorized workers for cleanup of small spills of acetone (about one litre) for a particular worksite:

(1) Extinguish and control all ignition sources, including electrical services, open flames, and electrostatic discharge.
(2) Evacuate workers to the designated safe location.
(3) Report the spill to your supervisor.
(4) Get the waste containers and spill cart.
(5) Put on the respirator, butyl rubber gloves, and safety goggles.
(6) Clean up the acetone using chemical absorbent pillows from the spill cart according to the manufacturer's instructions.
(7) Do not flush or rinse the spilled acetone into the sewer system.
(8) Place used absorbent pillows (containing acetone) in designated waste containers.
(9) Dispose of used chemical absorbent pillows according to local waste disposal procedures.

7 Tables and checklists

WHMIS 2015 information items on a supplier label

Information item	Description
1 Product identifier	The brand name, chemical name, common name, generic name, or trade name of the hazardous product.
2 Initial supplier identifier	The name, address, and telephone number of either the Canadian manufacturer or the Canadian importer.*
3 Pictogram(s)	Hazard symbol typically contained within a red, diamond-shaped border.
4 Signal word	A word used to alert the reader to a potential hazard and to indicate the severity of the hazard. "Danger" is used for high-risk hazards, while "Warning" is used for less severe hazards.
5 Hazard statement(s)	Standardized phrases that describe the nature of the hazard posed by a hazardous product. Examples include "Extremely flammable gas," "Fatal if inhaled," and "May cause cancer."
6 Precautionary statement(s)	Standardized phrases that describe measures to be taken to minimize or prevent adverse effects resulting from: • Exposure to a hazardous product, or • Improper handling or storage of a hazardous product Examples include "Keep container tightly closed," "Wear eye protection," and "If exposed or concerned: Get medical attention."

In rare cases, labels may include supplemental label information about precautionary actions, hazards not yet included in GHS, physical state, or route of exposure. This information must not contradict or detract from the standardized information. Supplemental label information is only included on a supplier label when:

- A toxic mixture has an ingredient with unknown acute toxicity, or
- A product reacts with water to produce an acutely toxic gas

* Initial supplier identifier — There are two exceptions to this requirement:

- In a situation where a hazardous product is being sold by a distributor, the distributor may replace the name, address, and telephone number of the initial supplier with their own contact information.
- In a situation where an importer imports a hazardous product for use in their own workplace in Canada (i.e., the importer is not selling the hazardous product), the importer may retain the name, address, and telephone number of the foreign supplier on the label and SDS instead of substituting their own contact information.

Information items on an SDS (full list)

SDS section and heading	Specific information elements
1 Identification	• Product identifier (e.g., product name) • Other means of identification (product family, synonyms, etc.) • Recommended use • Restrictions on use • Canadian supplier identifier[+] – Name, full address, and phone number(s) • Emergency telephone number and any restrictions on the use of that number, if applicable
2 Hazard identification	• Hazard classification (class, category) of substance or mixture, or a description of the identified hazard for Physical or Health Hazards Not Otherwise Classified • Label elements: – Pictogram or the name of the pictogram (e.g., flame; skull and crossbones) – Signal word – Hazard statement(s) – Precautionary statement(s) • Other hazards which do not result in classification (e.g., molten metal hazard)
3 Composition/ Information on ingredients	• When a hazardous product is a material or substance: – Chemical name – Common name and synonyms – Chemical Abstracts Service (CAS) registry number and any unique identifiers – Chemical name of impurities, stabilizing solvents and/or additives* • For each material or substance in a mixture that is classified in a health hazard class**: – Chemical name – Common name and synonyms – CAS registry number and any unique identifiers – Concentration **NOTE:** Confidential business information rules can apply.

SDS section and heading	Specific information elements
4 First-aid measures	• First-aid measures by route of exposure: – Inhalation – Skin contact – Eye contact – Ingestion • Most important symptoms and effects (acute or delayed) • Immediate medical attention and special treatment, if necessary
5 Fire-fighting measures	• Suitable extinguishing media • Unsuitable extinguishing media • Specific hazards arising from the hazardous product (e.g., hazardous combustion products) • Special protective equipment and precautions for firefighters
6 Accidental release measures	• Personal precautions, protective equipment, and emergency procedures • Methods and materials for containment and cleaning up
7 Handling and storage	• Precautions for safe handling • Conditions for safe storage (including incompatible materials)
8 Exposure controls/ Personal protection	• Control parameters, including occupational exposure guidelines or biological exposure limits and the source of those values • Appropriate engineering controls • Individual protection measures (e.g., personal protective equipment)

SDS section and heading	Specific information elements
9 Physical and chemical properties	• Appearance (physical state, colour, etc.) • Odour • Odour threshold • pH • Melting point / Freezing point • Initial boiling point / boiling range • Flash point • Evaporation rate • Flammability (solid; gas) • Lower flammable/explosive limit • Upper flammable/explosive limit • Vapour pressure • Vapour density • Relative density • Solubility • Partition coefficient — n-octanol/water • Auto-ignition temperature • Decomposition temperature • Viscosity
10 Stability and reactivity	• Reactivity • Chemical stability • Possibility of hazardous reactions • Conditions to avoid (e.g., static discharge, shock, or vibration) • Incompatible materials • Hazardous decomposition products
11 Toxicological information	Concise but complete description of the various toxic health effects and the data used to identify those effects, including: • Information on the likely routes of exposure (inhalation, ingestion, skin and eye contact) • Symptoms related to the physical, chemical, and toxicological characteristics • Delayed and immediate effects, and chronic effects from short-term and long-term exposure • Numerical measures of toxicity

SDS section and heading	Specific information elements
12 Ecological information***	• Ecotoxicity • Persistence and degradability • Bioaccumulative potential • Mobility in soil • Other adverse effects
13 Disposal considerations***	Information on safe handling for disposal and methods of disposal, including any contaminated packaging
14 Transport information***	• UN number • UN proper shipping name • Transport hazard class(es) • Packing group • Environmental hazards • Transport in bulk, if applicable • Special precautions
15 Regulatory information***	Safety, health, and environmental regulations specific to the product
16 Other information	Date of the latest revision of the SDS

+ The supplier that must be identified on an SDS is the initial supplier identifier (i.e., the name, address, and telephone number of either the Canadian manufacturer or the Canadian importer). There are two exceptions to this requirement. In a situation where a hazardous product is being sold by a distributor, the distributor may replace the name, address, and telephone number of the initial supplier with their own contact information. In a situation where an importer imports a hazardous product for use in their own workplace in Canada (i.e., the importer is not selling the hazardous product), the importer may retain the name, address, and telephone number of the foreign supplier on the SDS and label instead of substituting their own contact information.

* These impurities and stabilizing products are those that are classified in a health hazard class and contribute to the classification of the material or substance.

** Each ingredient in the mixture must be listed when it is classified in a health hazard class and:
  - Is present above the concentration limit that is designated for the hazard class in which it is classified, or
  - Is present in the mixture at a concentration that results in the mixture being classified in any health hazard class

*** Sections 12 to 15 require the headings to be present, but under Canadian regulations, the supplier has the option to not provide information in these sections.

WHMIS 2015 implementation plan checklist

Activity	Date needed	Assigned to	Date completed
Assign responsibility for WHMIS implementation			
1			
2			
3			
Establish an inventory of hazardous products			
Determine which products used or produced are classified as hazardous products under WHMIS.			
WHMIS labels and SDSs			
Develop labels and SDSs for hazardous products produced in the workplace.			
Obtain SDSs for hazardous products already in the workplace.			
Develop a process for requesting and receiving SDSs for new purchases.			
Develop methods to store SDSs so that they are readily available to workers.			
Develop a process to ensure that supplier labels are on or available for all new hazardous products received.			
Develop a process to ensure that information on SDSs and supplier labels is updated as soon as significant new data is provided.			
Develop a process to create and provide workplace labels and other means of identification.			

Activity	Date needed	Assigned to	Date completed
Determine hazards			
Identify and evaluate the hazards of hazardous products in the workplace (for example, consider the quantities to be used and stored, and the work processes where these products are used).			
Workplace controls			
Based on the hazard evaluation, determine where the following workplace controls may need to be established or upgraded: • Substitution of a less-hazardous product • Engineering controls such as local exhaust ventilation and process modification • Administrative controls such as work procedures and work scheduling • Personal protective equipment and clothing			
Integrate these controls into the overall health and safety program.			
Emergency procedures			
Review first aid procedures and upgrade them if required.			
Review spill control procedures and upgrade them if required.			
Review firefighting procedures and upgrade them if required.			
Notify the local fire department of the location, types, and quantities of hazardous products used and stored.			

Activity	Date needed	Assigned to	Date completed
Worker education and training			
Complete the "WHMIS 2015 education and training checklist" (page 86).			
Evaluate WHMIS program			
Establish a periodic review process for the following: • Check to ensure that no SDS is more than three years old. • Check that all required items on the SDS are present and have been completed. • Check the condition and presence of labels for all hazardous products. • Monitor workplace controls to ensure they are effective. • Review the WHMIS requirements, including education and training, at least annually.			

WHMIS 2015 education and training checklist

Activity	Assigned to	Date completed
Development		
Consult the occupational health and safety committee or worker representative on the development, implementation, and review of the program.		
Identify all hazardous products used in the workplace.		
Evaluate the hazards of each hazardous product.		
Identify WHMIS instructors, from either internal or external sources.		
Educate and train instructors (if internal), or evaluate their qualifications (if external).		
Identify employees to be educated and trained—those who work with or near hazardous products.		
Establish a process to identify new employees and contractors who require education and training.		
Evaluate labels and SDSs to be used in the education and training program (check for clarity, accuracy, and completeness).		
Evaluate safe work and emergency procedures to be used in the WHMIS education and training program.		
Education and training		
Provide a general introduction to WHMIS (for example, discuss responsibilities, labels, and SDSs).		
Provide instruction on how to identify hazardous products.		
Provide instruction on control measures and safe work procedures.		
Provide instruction on emergency procedures.		

Activity	Assigned to	Date completed
Provide instruction on accessing information on hazardous products.		
Evaluate the need for additional or specialized instruction to workers (for example, to those with language or learning difficulties), and provide this instruction where required.		
Provide instruction to workers whenever new products are received or new hazard information becomes available.		
Provide training on workplace-specific control measures and safe work procedures.		
Provide training on workplace-specific emergency procedures.		
Follow-up activities		
Evaluate workers' understanding of WHMIS using suitable means such as written tests and practical demonstrations. Provide further education and training as required.		
Review the effectiveness of the education and training program at least once a year. (Reviews must be done in consultation with the occupational health and safety committee or the worker representative.)		

8 Resources

WorkSafeBC resources

WorkSafeBC produces a variety of WHMIS publications and videos, many of which are available online at no charge (see links below). Hard copy versions of some of these resources can be ordered from the WorkSafeBC Store:

Phone: 604.232.9704
Toll-free phone: 1.866.319.9704
Fax: 604.232.9703
Toll-free fax: 1.888.232.9714
Online ordering: worksafebcstore.com

WHMIS information

For links to a wide range of WHMIS information, visit worksafebc.com/en/health-safety/hazards-exposures/whmis

Occupational Health and Safety Regulation and OHS Guidelines

Available online at worksafebc.com.
worksafebc.com/en/law-policy/occupational-health-safety/searchable-ohs-regulation

- Part 5, Chemical Agents and Biological Agents, Workplace Hazardous Materials Information System (WHMIS)
  worksafebc.com/en/law-policy/occupational-health-safety/searchable-ohs-regulation/ohs-regulation/part-05-chemical-and-biological-substances#F90B861D458E46028AAE4F574CC2CC7C
- Guidelines G5.3-1 to G5.15, Workplace Hazardous Materials Information System (WHMIS)
  worksafebc.com/en/law-policy/occupational-health-safety/searchable-ohs-regulation/ohs-guidelines/guidelines-part-05#7BE29C5F4B2942A58E5B8E037C6D9B7D

Videos

WHMIS 2015 for Employers
worksafebc.com/en/resources/health-safety/videos/whmis-2015-for-employers
In less than six minutes, this video explains the primary changes from WHMIS 1988 to WHMIS 2015: new hazard classes, new labelling requirements, and a revised safety data sheet format. It also describes your responsibilities as an employer.

WHMIS 2015 for Workers
worksafebc.com/en/resources/health-safety/videos/whmis-2015-for-workers
This four-minute video helps workers understand the primary changes from WHMIS 1988 to WHMIS 2015 — new hazard classes, new labelling requirements, and a revised safety data sheet format — as well as the responsibilities of their employers to provide education and training.

Publications
WHMIS 2015: The Basics
worksafebc.com/en/resources/health-safety/whmis/whmis-basics-2015
By understanding the information in this book, workers learn where to look on the labels of hazardous products, where to look on safety data sheets, and what to discuss with their supervisors. Workers also learn about key changes from WHMIS 1988 to WHMIS 2015.

Other sources of information

Organizations

Contact the following organizations for publications mentioned in this book or for other relevant WHMIS publications.

Canadian Centre for Occupational Health and Safety
135 Hunter Street East
Hamilton ON L8N 1M5
Phone: 1.800.668.4284
Website: ccohs.ca
The CCOHS provides a wide range of health and safety information for both workers and employers. Examples of WHMIS 2015 training resources include the *WHMIS 2015 Instructor's Toolkit* and the *WHMIS 2015 Participant Workbook*.

Workplace Hazardous Materials Bureau of Health Canada
269 Laurier Avenue West, 7th Floor (4903E)
Ottawa, Ontario K1A 0K9
Telephone: 1.855.407.2665
Teletypewriter: 1.800.465.7735 (Service Canada)
Website: canada.ca/en/health-canada/corporate/contact-us/workplace-hazardous-materials-information-system.html
Email: WHMIS_SIMDUT@hc-sc.gc.ca
The Workplace Hazardous Materials Bureau of Health Canada coordinates the administration of WHMIS and serves as the coordinator for the governance and administration of WHMIS in Canada. Health Canada also acts as the secretariat for this federal, provincial, and territorial government partnership system. In addition, Health Canada handles claims for exemptions for confidential business information (CBI).

Commission des normes, de l'équité, de la santé et de la sécurité du travail
Répertoire toxicologique
1199, rue de Bleury, 3e étage
CP 6056, SUCC Centre-Ville
Montréal QC H3C 4E1
Phone: 514.906.3080, poste 2304
Website: reptox.csst.qc.ca
The CNESST's toxicological directory provides information on the classification of chemicals and health and safety risks.